Pattern Recognition 170 (2026) 112010

Contents lists available at ScienceDirect

Pattern Recognition

journal homepage: www.elsevier.com/locate/pr

Bidirectional skip-frame prediction for video anomaly detection with intra-domain disparity-driven attention

Jiahao Lyu [a], Minghua Zhao [a,*], Jing Hu [a], Runtao Xi [b,c], Xuewen Huang [a], Shuangli Du [a], Cheng Shi [a], Tian Ma [d]

[a] School of Computer Science and Engineering, Xi'an University of Technology, Xi'an, China
[b] CCTEG Changzhou Research Institute, Changzhou, China
[c] Tiandi (Changzhou) Automation Co., Ltd., Changzhou, China
[d] College of Computer Science and Technology, Xi'an University of Science and Technology, Xi'an, China

ARTICLE INFO

Keywords:
Video anomaly detection
Bidirectional frame prediction
Intra-domain disparity
Attention mechanism
Autoencoder

ABSTRACT

Unsupervised video anomaly detection (VAD) trains only normal events to detect anomalies by large discrepancies during testing. Expanding the discriminative boundary between normal and abnormal events is the common goal and challenge of unsupervised VAD. To address this problem, we propose a Bidirectional Skip-frame Prediction (BiSP) method, leveraging intra-domain disparities between different features. Specifically, the BiSP skips frames in the training phase to achieve the forward and backward prediction respectively, and in the testing phase, it utilizes bidirectional consecutive frames to co-predict the same intermediate frames. Furthermore, we design the variance channel attention and context spatial attention from the perspectives of movement patterns and object scales, respectively. This design maximizes of the disparity between normal and abnormal in the feature extraction and delivery with different dimensions. Extensive experiments from four benchmark datasets demonstrate the effectiveness of the proposed BiSP, which substantially outperforms state-of-the-art competing methods. Ours code is available at https://github.com/jLooo/BiSP.

1. Introduction

With advancements in computer vision, anomaly detection has been widely researched across various fields and applications, such as medical image anomaly detection [1], industrial quality inspection [2], and traffic surveillance [3]. In particular, with the widespread deployment and application of video surveillance equipment across various fields play an important role in guaranteeing social security and stability. Simultaneously, the quantity of surveillance videos has shown explosive growth, and manual detection methods cannot meet the demand for timely and accurate detection of massive video. Therefore, intelligent technology is gradually replacing manual detection, among that, video anomaly detection (VAD) has been investigated and studied by researchers with the development of computer vision.

VAD can be categorized into supervised learning [4], weakly-supervised learning [5], and unsupervised learning [6]. Supervised VAD methods are not the primary method because they need to utilize the fine-grained label or image-level label data to complete the discrimination between normal and anomaly. The anomaly events are usually discrete and occur with low probability in the video sequence, making detection time-consuming and inefficient. Similarly, weakly-supervised VAD methods utilize image-level labels for anomaly detection, and the training dataset consists of labeled normal and anomaly events. These methods differentiate between normal and anomaly through multi-instance learning ranking [7] or classification models [8]. Compared to the aforementioned methods, unsupervised VAD methods do not require any labels. Instead, they only need to complete the classification preprocessing of normal and abnormal events when constructing the dataset, i.e., only modeling and feature learning of normal events in the training phase, and completing the anomaly detection by measuring the reconstruction deviation of the abnormal events in the testing phase. This approach can simply locate the specific temporal frames and spatial locations of the abnormal events. Unsupervised VAD methods follow the generalized assumption [9] for anomaly detection tasks, namely, that any event not seen in the training set is considered anomalous. Therefore, the boundary between normal and abnormal events is critical. However, a major challenge arises when this boundary is ambiguous [9], the anomaly detection model will suffer tremendously, leading to performance degradation. To this end,

* Corresponding author.
E-mail addresses: jhlyu25@stu.xaut.edu.cn (J. Lyu), zhaominghua@xaut.edu.cn (M. Zhao), jinghu@xaut.edu.cn (J. Hu), jusadw@163.com (R. Xi), xuewenhuang@stu.xaut.edu.cn (X. Huang), dusl@xaut.edu.cn (S. Du), chengc_s@163.com (C. Shi), matian@xust.edu.cn (T. Ma).

https://doi.org/10.1016/j.patcog.2025.112010
Received 11 August 2024; Received in revised form 19 April 2025; Accepted 9 June 2025
Available online 23 June 2025
0031-3203/© 2025 Published by Elsevier Ltd.

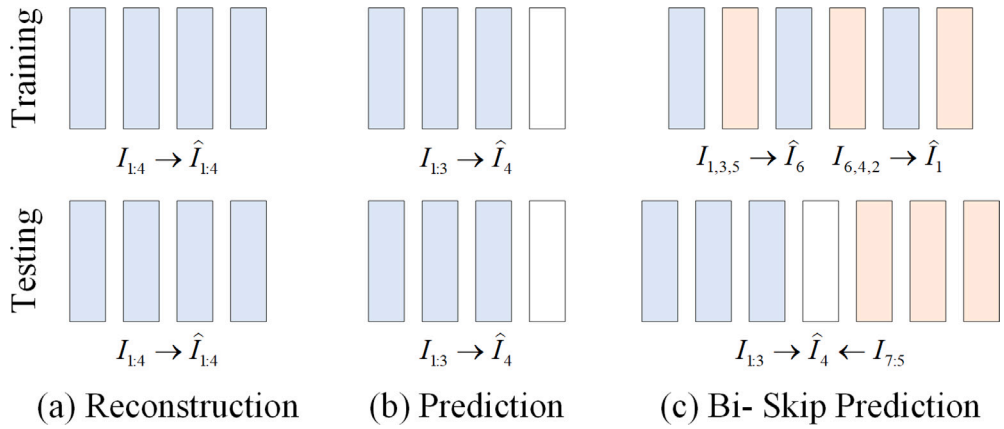

Fig. 1. Three types of video anomaly detection methods. (a) Reconstruction-based method. (b) Prediction-based method. (c) The proposed Bidirectional Skip-frame Prediction method.

previous works have explored spatio-temporal feature extraction [10] and attention mechanisms [11], among which, most of the methods implement reconstruction-based or prediction-based methods with the Autoencoder (AE) framework, where the encoder compresses the input features into low-dimensional representations, and the decoder reconstructs these compressed features, using reconstruction or prediction errors to complete VAD. Compared to reconstruction-based methods, prediction-based methods have been widely researched as they can model both temporal and spatial features. As shown in Fig. 1(a)(b), reconstruction-based methods input and output are the same frames, i.e., input continuous frames $I_{1:4}$, output reconstructed frames $\hat{I}_{1:4}$. Whereas the prediction-based methods input $I_{1:3}$, output the predicted frame $\hat{I}_4$.

In the current reconstruction [6] or prediction methods [12], the anomaly is determined by calculating the reconstruction/prediction error between the original frame and the reconstructed/predicted frame. For normal events, the reconstruction/prediction error should be small, while for abnormal events, it should be large. Therefore, how to minimize the normal errors and maximize the anomaly errors is the essential problem of VAD. However, most methods only determine the detection accuracy by a single type of error. To improve detection comprehensiveness and reduce the false alarm rate, some works have proposed hybrid anomaly detection [13] or bidirectional frame prediction [14] methods, such as fusing the frame error with the optical flow error, fusing the reconstruction error with the prediction error, and fusing the forward prediction error with the backward prediction error, thereby enhancing detection performance from different perspectives. Therefore, some methods utilize preprocessing operations such as the optical flow extraction [15] or the foreground mask [16] as auxiliary information, but the added computational cost makes them unable for real-time tasks.

Due to the weak intra-domain gap between normal events and the large intra-domain disparity between normal and anomaly events, the VAD method allows normal events to form clusters in the feature space, while most abnormal features fail to belong to the same cluster with them, or even form separate abnormal clusters. Inspired by this and following the bidirectional prediction method [14,17], this paper designs a bidirectional skip-frame prediction (BiSP) method, as shown in Fig. 1(c). Unlike traditional single-frame [12] and dual-frame [14] prediction methods, the proposed training strategy uses skip frames to predict future frames at both ends of the bidirectional direction. For instance, using forward skip frames $I_{1,3,5}$ to predict $\hat{I}_6$ and using backward skip frames $I_{6,4,2}$ to predict $\hat{I}_1$. Compared to traditional prediction method, the proposed method more effectively extracts the motion features more easily and focuses on the normal events with the weak intra-domain gap by the proposed attention. In the testing phase, the prediction mode is changed to further increase the intra-domain disparity between the normal and abnormal events. Ultimately, we achieve intra-domain disparity-driven VAD for surveillance videos. Specifically, based on the dual-stream AE framework, we construct a variance channel attention with parallel structure and a context spatial attention with serial structure, which enhances the model's ability to discriminate the normality of features of different dimensions during the skip connection and decoding processes. On the one hand, variance channel attention enhances the ability of the model to discriminate movement patterns between different events. On the other hand, context spatial attention enriches the feature extraction and representation ability of the model for objects with different scales. In the testing phase, consecutive frames at both ends of the input are used to co-predict the same intermediate frames, and the disparity between normal and abnormal events is enlarged compared to the weak gap between normal and normal events.

In brief, the proposed method leverages attention mechanisms to expand the discrimination boundary between normal and abnormal events in both forward and backward prediction, and finally achieve efficient detection by maximum anomaly error. To summarize, the main contributions of this paper are as follows:

- We propose a novel bidirectional skip-frame prediction method that employs the dual-AE structure for both forward and backward prediction. Compared to state-of-the-art methods, our method demonstrates competitive performance on benchmark datasets.
- Leveraging the inherent intra-domain disparity between normal and anomalous events, the proposed method further amplifies this disparity by employing distinct video frame input strategies during the training and testing stages.
- To enhance feature representation, we propose variance channel attention and context spatial attention respectively. These two attention mechanisms are designed to improve discrimination of different anomaly categories, focusing on movement patterns and appearance characteristics, respectively.

The rest of this paper is organized as follows: Section 2 discusses the related work for VAD, Section 3 introduces the framework of the proposed BiSP, Section 4 presents the experimental and visualization results on the benchmark dataset, Section 5 discuss the limitations and the future work and Section 6 concludes the paper.

2. Related work

The majority of video clips captured by surveillance equipment depict normal events, making abnormal events even rarer and harder to

define. Therefore, unsupervised learning is generally more reliable for VAD. Unsupervised VADs are typically categorized into reconstruction-based [6] and prediction-based [12], both of which discriminate abnormal events by measuring the error in the reconstructed or predicted frames. The difference is that reconstruction-based methods identify errors across multiple consecutive frames corresponding to the input, while prediction-based methods focus on errors in future frames. Most prediction methods predict a single frame for the next moment, while a few methods [3,18] still opt for dual-frame prediction.

Reconstruction-based Methods. Reconstruction methods focus on reconstructing the spatial features of the normal samples due to the input and output of the frames at the same time. For instance, Gong et al. [6] proposed a memory-augmented AutoEncoder (memAE), which embeds an external memory network between the encoder and decoder to record the feature distributions of normal events for efficient reconstruction. Lappas et al. [19] introduced Dynamic Distinction Learning (DDL), which enhances VAD accuracy through the fusion of pseudo anomaly and dynamic anomaly weighting. Kommanduri et al. [20] introduce the DenserResNet AutoEncoder, which utilizes multiple dense and residual networks to enhance contextual understanding of features at different scales. Park et al. [21] proposed an attention-based memory addressing mechanism, called MNAD, which updates the memory pool during the testing phase to ensure that the model can identify normal events. In the same framework, MNAD based on prediction outperforms reconstruction, as similarly shown in [22]. This is because reconstruction methods cannot model the temporal feature dimension and ignore the temporal relationship between frames.

Prediction-based Methods. In contrast, the output frames of prediction methods are different from the input frames, which allows better modeling and discovery of temporal relationships in consecutive input frames. Frame-Pred, proposed by Liu et al. [12], is the first prediction framework for VAD. Huang et al. [23] introduced a novel approach to produce rotation-detectable for normal frames, which improves the discriminative ability of the self-supervised discriminator by reducing its generalization to anomaly frames. Ning et al. [24] proposed the Memory-Augmented Appearance-Motion Consistency framework, applying channel attention to multi-scale features to capture the interaction between appearance and motion (optical flow), enhancing the robustness and effectiveness of the model. Huang et al. [25] proposes a novel variational normal inference (VNI) model to evaluate the distribution of potential features of normal events, as well as a marginal learning embedding module to optimize VNI model for training. It is shown to remain able to achieve good performance in cross-validation datasets. Unlike 2D convolution, 3D convolution enables joint extraction of temporal and spatial features. Park et al. [10] designed a spatio-temporal patch transformation model, where consecutive frames are chopped up in the training phase and made into a frame rectangle as input and randomly generates patch anomalies within the rectangle, which aids feature learning. Hao et al. [26] proposed a spatio-temporal coherence enhancement network that combines motion and image appearance through 3D-2D convolutional architecture. The 3D convolution extract spatio-temporal high-level features during encoding, and then feature downscaling is achieved through resampling. Finally, the prediction is completed using 2D convolution during the decoding process. Qiu et al. [22] enhanced the ability of the model to learn spatio-temporal features by extracting I3D features, and designed a dual-scale feature cluster module at latent space to expand the discriminative boundary between normal and abnormal events. However, the 3D convolution in the above methods is computationally expensive and can impact detection efficiency.

Furthermore, hybrid detection and dual-frame prediction methods have demonstrated superior anomaly detection performance. MAAM-Net [27] incorporates a memory augmented module for both single-frame reconstruction and optical flow prediction, enabling error fusion across these two tasks. Fang et al. [14] proposed SIamese generative network (SIGnet), which reformulates anomaly detection as a mutually supervised problem. It uses the two identical generators to complete the prediction of the same video frame between two different video frames, and employs a prediction consistency loss function, thus improving the model generalization ability. Li et al. [3] proposed a multi-branch generative adversarial network with context learning (MGAN-CL) method that learns the context information to determine whether abnormal events occur. Zhang et al. [18] utilizes mutual learning and inspiration between different results by using two generators with identical structure but different initialization to predict the same future frames. This approach allows the generators to learn from each other, indirectly increasing the diversity of the training samples, reducing model overfitting to normal events, and enhancing anomaly detection accuracy. Zhong et al. [17] based on bidirectional prediction with the CBAM [28], which enables the forward and backward prediction models to achieve high-quality bidirectional detection by adaptively fusing bidirectional spatio-temporal features, respectively.

Compared to reconstruction-based methods, prediction-based methods achieve better detection performance by well-modeling temporal features in the samples. Such methods can better capture the dynamic changes by considering the temporal and spatial relationships between different events. In contrast, reconstruction-based methods, which do not account for the temporal relationships among the samples, may struggle to recognize normal and abnormal events, often reconstructing both types of events together. To address this challenge, and inspired by the intra-domain disparity between normal and abnormal events, this paper proposes a new prediction method with two attention mechanisms.

3. Method

In this section, we introduce a novel unsupervised anomaly detection prediction method, BiSP, and analyze its objectives. Meanwhile, the variance channel attention and context spatial attention mechanisms are designed to enhance the model's ability to determine the disparity between normal and anomaly events during different feature delivery processes, respectively. The framework of the proposed BiSP is illustrated in Fig. 2(a).

3.1. Bidirectional skip-frame prediction framework

From the perspective of expanding the intra-domain disparity between normal and abnormal events, we propose the BiSP, which consists of three modules: dual-stream AE, variance channel attention (VarCA), and context spatial attention (ConSA). Fig. 2(b) shows the forward frame prediction network in BiSP, which first selects a continuous video frame, then divides it into two separate skip frames: forward skip frames I_{fwd} and backward skip frames I_{bwd}. These skip frames are fed into the corresponding forward $P_f = \{E_f, D_f\}$ and backward $P_b = \{E_b, D_b\}$ AE to construct the bidirectional skip-frame prediction network, where E and D denote the encoder and decoder, respectively. The dual AE are identical in structure, differing only in how they extract forward or backward temporal features. Forward prediction network $P_f(I_{fwd}) \rightarrow \hat{I}_f$: input forward skip frames I_{fwd}, output forward predicted frame $\hat{I}_f$. Correspondingly, the backward prediction network $P_b(I_{bwd}) \rightarrow \hat{I}_b$: input backward skip frames I_{bwd}, output backward predicted frame $\hat{I}_b$. Table 1 shows the detailed composition and specific parameters of the encoders and decoders in the single AE network structure, VarCA and ConSA will be discussed in the subsequent section.

BiSP aims to perform the bidirectional prediction of frame segments based on bidirectional skip features. To achieve this, BiSP uses the L2 distance to compute the corresponding ground truth errors as forward prediction loss $\mathcal{L}_{fp}$ and backward prediction loss $\mathcal{L}_{bp}$:

$$\mathcal{L}_{fp} = \left\| \hat{I}_f - I_f \right\|_2^2, \tag{1}$$

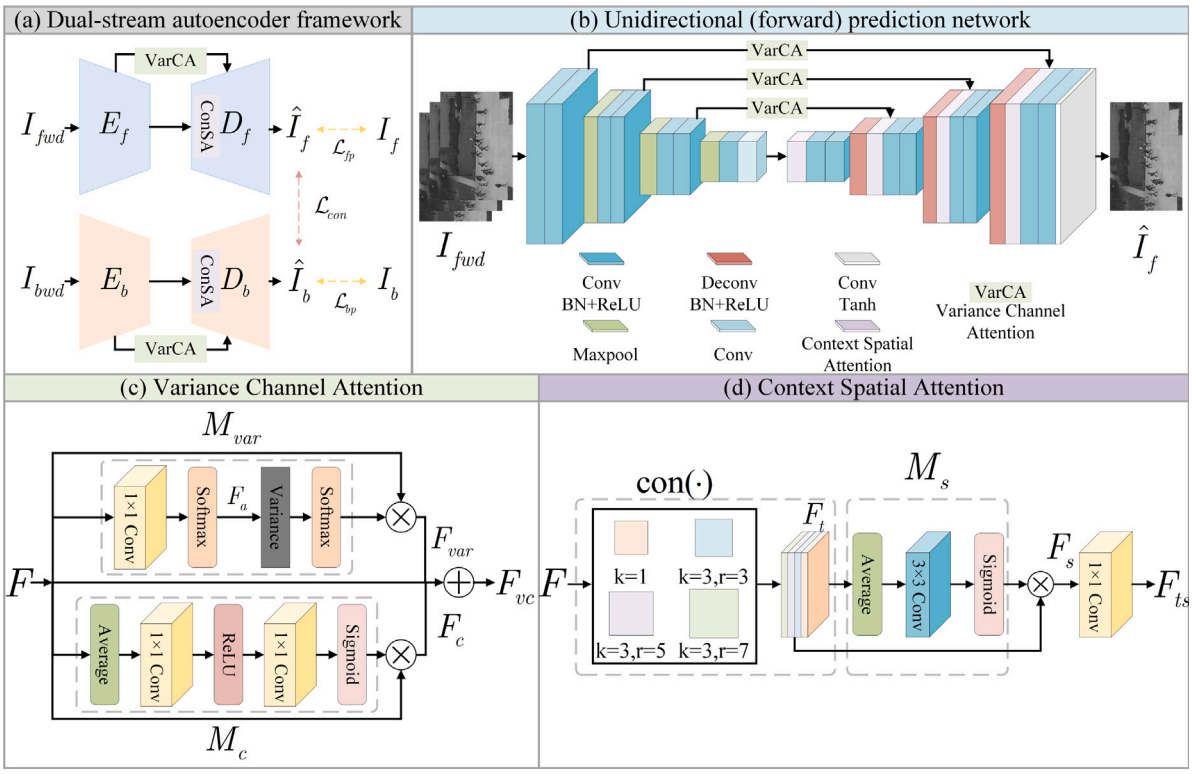

Fig. 2. Overview of the bidirectional skip-frame prediction framework. (a) We design two independent AEs to extract forward and backward skip frame features and predict the next frame respectively. (b) The dual AE have the same structure, so the forward prediction network is used as an example. (c) The parallel structure for variance channel attention. (d) The serial structure for context spatial attention.

Table 1
The detail of single AE network structure.

Stage	Layers	Kernel size	Stride
Encoder	[Conv+BN+ReLU] ×2	3 × 3 × 32	1
	[Maxpool]	2 × 2 × 32	2
	[Conv+BN+ReLU] ×2	3 × 3 × 64	1
	[Maxpool]	2 × 2 × 64	2
	[Conv+BN+ReLU] ×2	3 × 3 × 128	1
	[Maxpool]	2 × 2 × 128	2
	[Conv+BN+ReLU]	3 × 3 × 256	1
	[Conv]	3 × 3 × 256	1
Decoder	[Conv+BN+ReLU] ×2	3 × 3 × 256	1
	[DeConv+BN+ReLU]	3 × 3 × 128	2
	[Conv+BN+ReLU] ×2	3 × 3 × 128	1
	[DeConv+BN+ReLU]	3 × 3 × 64	2
	[Conv+BN+ReLU] ×2	3 × 3 × 64	1
	[DeConv+BN+ReLU]	3 × 3 × 32	2
	[Conv+BN+ReLU] ×2	3 × 3 × 32	1
	[Conv+Tanh]	3 × 3 × 3	1

$$\mathcal{L}_{bp} = \left\| \hat{I}_b - I_b \right\|_2^2. \tag{2}$$

Due to the differing prediction goals in the training and testing phases, to ensure the structural consistency between the two final prediction frames in the testing phase, we introduce the Structural Similarity Index Measure (SSIM) to construct the consistency loss $\mathcal{L}_{con}$ of the two prediction frames:

$$\mathcal{L}_{con} = 1 - \mathrm{SSIM}(\hat{I}_f, \hat{I}_b). \tag{3}$$

Ultimately, the combination of loss function including two prediction losses and consistency loss is shown in Eq. (4):

$$\mathcal{L} = \mathcal{L}_{fp} + \mathcal{L}_{bp} + \mathcal{L}_{con}. \tag{4}$$

VAD methods typically use anomaly scores to determine whether each video frame is abnormal or not. Most methods employ peak signal-to-noise ratio (PSNR) to measure the anomaly scores $S(I_t)$, where PSNR is calculated by the mean-square error $e(I_t, \hat{I}_t)$ between the ground truth I_t and the predicted frame $\hat{I}_t$. Since two identical frames are predicted in the testing phase of this work, we thus fused the forward $e_f(I_f, \hat{I}_f)$ and backward $e_b(I_b, \hat{I}_b)$ errors by the weighted sum strategy, as shown in Eq. (5):

$$e(I_t, \hat{I}_t) = w_f \cdot e_f + w_b \cdot e_b \tag{5}$$

where w_f and w_b are the weights of the e_f and e_b scores respectively. Because the two predictions of BiSP are essentially identical frames, we regard them as complementary to each other, and thus w_f and w_b sum to one ($w_f + w_b = 1$).

Based on the fusion error, we use the multi-scale anomaly evaluation method [17] for anomaly score calculation, which can be summarized as achieving a more comprehensive detection of anomalies at three different scales ($N = 3$) through the error pyramid, as shown in Eq. (6):

$$\mathrm{PSNR}(I_t, \hat{I}_t) = 10 \log_{10}\left(\frac{1}{\sum_{i=0}^{N} v_i}\right), \tag{6}$$

where v_i is the maximum prediction mean-square error based on the patch block in scale i obtained by mean pooling. Finally, the PSNR is normalized to $[0, 1]$ by Eq. (7), and smoothed by a Gaussian filter to produce the final anomaly score.

$$S(I_t) = \frac{\mathrm{PSNR}(I_t, \hat{I}_t) - \min(\mathrm{PSNR}(I_t, \hat{I}_t))}{\max(\mathrm{PSNR}(I_t, \hat{I}_t)) - \min(\mathrm{PSNR}(I_t, \hat{I}_t))}. \tag{7}$$

3.2. Variance channel attention

The appearance disparity between most anomaly events and normal events is relatively significant, allowing methods to easily cope with the

above situations and achieve high anomaly scores. However, for events with ambiguous boundaries, many methods struggle to effectively handle such cases. For this reason, we propose a parallel structure for the variance channel attention (VarCA) mechanism, as shown in Fig. 2(c). Because part of the anomalies are related to movement patterns, VarCA increases the feature variance between different events. During the testing phase, the variance and channel attention focus on motion gaps and features of normal events, which increases the prediction errors for abnormal events. Specifically, the variance of the spatial dimensions $D_s = H \times W$ across different channels C is computed after modeling the global feature $F_a \in \mathbb{R}^{B\times D_s\times 1}$ using a 2D convolution and a softmax activation function:

$$F_a = \mathrm{softmax}(W_1^1(F)), \tag{8}$$

where W_1^1 denotes the 2D convolution with 1×1 filter sizes and 1 channel. Then, the input feature $F \in \mathbb{R}^{B\times C\times H\times W}$ is multiplied element-by-element with the results of variance attention map M_{var}, resulting in the variance feature $F_{var} \in \mathbb{R}^{B\times C\times H\times W}$:

$$M_{var}(F) = \mathrm{softmax}\left(\left\| F_a - \frac{1}{D_s}\sum_{d=1}^{D_s} F_a^d \right\|_2^2\right), \tag{9}$$

$$F_{var} = F \otimes M_{var}(F),$$

Similarly, the input feature F is also multiplied element-by-element with the channel attention map M_c, resulting in the channel feature $F_c \in \mathbb{R}^{B\times C\times H\times W}$:

$$\begin{aligned} M_c(F) &= \delta W_1^C(\sigma W_1^{C/2}(P_{avg}(F))), \\ F_c &= F \otimes M_c(F), \end{aligned} \tag{10}$$

where δ and σ denote the Sigmoid and ReLU activation functions, respectively. P_{avg} denotes the global average-pooling operation. W_1^C denotes the 2D convolution with 1×1 filter size and C channels, and $W_1^{C/2}$ so on.

Finally, the two attention features are added element-by-element with the input feature F, resulting in the variance channel feature $F_{vc} \in \mathbb{R}^{B\times C\times H\times W}$. The overview of VarCA is summarized as shown in Eq. (11):

$$F_{vc} = F + F_{var} + F_c \tag{11}$$

3.3. Context spatial attention

Extracting spatio-temporal features is crucial for anomaly detection, and prediction methods outperform reconstruction because they predict unseen video frames with certain temporal feature. Moreover, different event objects often have large variations in scale and location features, and the above features cannot be effectively extracted using basic convolution and pooling operations. Therefore, we propose a serial structure for context spatial attention (ConSA) mechanism, as shown in Fig. 2(d). We first design a context feature extraction module $\mathrm{con}(\cdot)$, which employs four 2D dilation convolutions with different kernel sizes k and expansion rates r to extract the features and splice them into context feature $F_t \in \mathbb{R}^{B\times C_t\times H\times W}$, and $C_t = 32 \times 4$ denotes the sum of the channels of the four dilation convolutions:

$$F_t = \mathrm{con}(F), \tag{12}$$

Then, the context feature F_t is multiplied element-by-element with the results of spatial attention map M_s to get the spatial feature $F_s \in \mathbb{R}^{B\times C_t\times H\times W}$:

$$\begin{aligned} M_s &= \delta(W_3^1(P_{avg}(F_t))), \\ F_s &= F_t \otimes M_s(F_t), \end{aligned} \tag{13}$$

where W_3^1 denotes the 2D convolution with 3×3 kernel sizes and 1 channel.

Table 2
The statistics of four unsupervised VAD datasets.

Dataset	Train Frames/Clips	Test Frames/Clips	Scenes	Anomalies	Example
Ped1	6,800/34	7,200/36	1	61	Biking
Ped2	4,560/16	2,550/12	1	21	Driving
Avenue	15,328/16	15,324/21	1	77	Throwing
Sh-Tech	274,515/330	42,883/107	13	158	Loitering

Finally, input to the 2D convolution with the same channels as the input feature $F \in \mathbb{R}^{B\times C\times H\times W}$ to get the final feature $F_{ts} \in \mathbb{R}^{B\times C\times H\times W}$. The overview of ConSA is summarized as shown in Eq. (14):

$$F_{ts} = W_1^C(F_s), \tag{14}$$

where W_1^C denotes a 2D convolution with 1×1 filter sizes, maintaining the same number of channels as F.

4. Experiments

4.1. Datasets and implementation details

To evaluate the qualitative and quantitative results of the proposed BiSP and to compare it with state-of-the-art algorithms, we conduct experiments on four benchmark unsupervised video anomaly detection datasets, Table 2 summarizes the statistics of these VAD datasets.

UCSD Ped1 & Ped2: The Ped1 & Ped2 [29] are the earliest proposed datasets. The Ped1 dataset contains a total of 70 video clips divided into 34 training clips and 36 testing clips, but due to the low pixel count of the video frames in Ped1, most VAD methods do not consider it as a benchmark dataset for the experiments. The Ped2 dataset contains 16 training clips and 12 testing clips, and its scenes are simple, and clear and have more distinct boundaries between normal and abnormal events, and the abnormal events are mostly biking, skateboarding, and driving, which are objects that are more distinct from humans.

CUHK Avenue: The Avenue [30] is a large-scale single-scene VAD dataset containing 16 training clips and 21 testing clips, with a total of 30,000 video frames and a large number of abnormal events related to human behavior, such as running, throwing a bag, and wrong direction.

ShanghaiTech Campus: The ShanghaiTech [31] (Sh-Tech) dataset contains 13 scenes, more than 150 abnormal events such as running, loitering, jumping forward, etc., and up to 300,000 frames of surveillance video, making it the most challenging unsupervised VAD dataset.

We conduct a large number of experiments using area under the curve (AUC) as the main evaluation metric. The proposed BiSP is trained using an Adam optimizer with a learning rate of 0.0002 and a cosine annealing scheduler to adjust the learning rate. The number of input frames t is empirically set to 6 in the training phase and 7 in the testing phase. The resolution of all input images is rescaled to 256×256, and the pixel values are normalized to the range of $[-1, 1]$. Following the assumptions of the unsupervised VAD [9], all training sets are composed of video clips consisting of normal events collected from real scenes, whereas video clips containing anomaly events are defined as testing sets. The error weights w_f and w_b for Ped1, Ped2, Avenue, and Sh-Tech are set to (0.3, 0.7), (0.5, 0.5), (0.1, 0.9) and (0.7, 0.3), respectively.

4.2. Comparison with state-of-the-art methods

We compare the proposed BiSP with different AE-based reconstruction (Recon.) and prediction (Pred.) methods on four benchmark datasets. The AUC performances of the different methods are shown in Table 3. It can be seen that BiSP outperforms the other methods on both Ped1, Ped2, and Sh-Tech datasets. Fig. 3 shows the frame-level

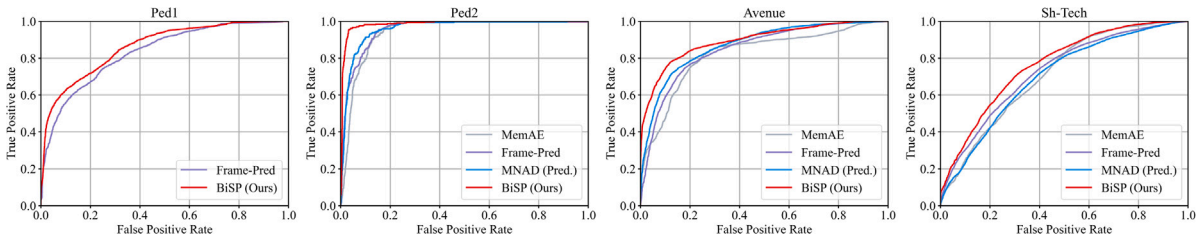

Fig. 3. ROC curves of different methods on four benchmark datasets.

Table 3
Comparative results of frame-level AUC (%) for the benchmark datasets. Best and second best performance are highlighted in bold and underlined, respectively.

	Method	Reference	Ped1	Ped2	Avenue	Sh-Tech
Recon.	MemAE [6]	ICCV19	–	94.1	83.3	71.2
	MNAD [21]	CVPR20	–	90.2	82.8	69.8
	Cluster-AE [32]	ECCV20	–	96.5	86.0	73.3
	MAAM-Net [27]	PR23	–	97.7	**90.9**	71.3
	Qiu et al. [22]	PR24	–	90.8	83.1	73.3
	C3DSU with DDL [19]	CVPR24	–	98.5	90.4	74.3
	DAST-Net [20]	NC24	85.4	97.9	89.8	73.7
Single-Pred.	Frame-Pred [12]	CVPR18	83.1	95.4	85.1	72.8
	MNAD [21]	CVPR20	81.1	97.0	88.5	70.5
	STCEN [26]	PR22	82.5	96.9	86.6	73.8
	SSAGAN [23]	TNNLS22	84.2	96.9	88.8	74.3
	DLAN-AC [33]	ECCV22	–	97.6	89.9	74.7
	Zhang et al. [11]	TCSVT22	85.2	95.8	84.9	71.4
	ASTNet [34]	APIN23	–	97.4	86.7	73.6
	MSN-net [35]	ICASSP23	–	97.6	89.4	73.4
	VADNet [25]	TCSVT23	–	96.8	87.3	75.2
	Qiu et al. [22]	PR24	–	92.2	86.2	73.8
	FastAno++ [36]	PR24	–	98.1	87.8	75.2
Dual-Pred.	MLIPN [18]	PRCV22	83.9	96.0	85.9	73.1
	SIGnet [14]	TNNLS22	86.0	96.2	86.8	–
	DEDDnet (Fusion) [17]	TCSVT22	–	98.1	89.0	74.5
	DEDDnet (Parallel) [17]	TCSVT22	–	97.4	88.6	–
	MGAN-CL [3]	TCSVT23	–	96.5	87.1	73.6
	BiSP (Ours)		**86.3**	**98.6**	89.5	**76.4**

Table 4
Comparisons of AUC (%) performances with state of-the-art methods that use object-level images as inputs.

Method	Reference	Object	Ped2	Avenue	Sh-Tech	FPS
HF2-VAD [15]	ICCV21	✓	99.3	91.1	76.2	10
SSMTL [40]	CVPR21	✓	98.7	93.7	82.9	–
BG-agnostic [41]	TPAMI22	✓	98.7	92.3	82.7	18
SSMTL++ [37][a]	CVIU23	✓	–	91.6	83.8	18
STG-NF [38][a]	ICCV23	✓	–	–	85.9	–
HSTGCNN [39][a]	TCSVT23	✓	97.7	87.5	81.8	14
BiSP(Ours)		✗	98.6	89.5	76.4	45

[a] Indicates the input includes pose data.

ROC curves comparing BiSP with the baseline method, where a larger area under the curve indicates a higher detection accuracy.

Reconstruction methods have outputs that are consistent with the inputs, thus making it difficult to learn temporal features, causing the performance of such methods to fall behind most prediction methods. However, we find that a few reconstruction methods [19,27] perform well in Avenue. This is because there are phenomena such as uneven background illumination in Avenue, which leads to the prediction methods not being able to focus on generating high-quality prediction frames, while the reconstruction methods can cope with this problem well. This paper focuses on comparing prediction methods, which are further categorized into single-frame prediction (Single-Pred.) and dual-frame prediction (Dual-Pred.).

Compared with the prediction methods, in Ped2, the boundary between normal and anomalies is sufficiently obvious, and the proposed BiSP can further expand the boundary of different events on this basis, thus improving the detection performance. Among the dual-frame prediction methods, MLIPN [18] achieves two results for unidirectional frame prediction through two independent generators, and the remaining four methods are bidirectional frame prediction, whose testing phases are to co-predict the same intermediate frames through the video frames at both ends, which is considered to be a simple and efficient detection way. SIGnet [14] and MGAN-CL [3] both use the AE as the generator, which is a much easier and more efficient detection mode than using only the AE as the primary network. GAN with better feature representation than DEDDnet [17] and BiSP which only use AE as the primary network. However, this is because DEDDnet and BiSP enhance the model's attention to normal events by designing different attention mechanisms, thus guaranteeing that some of the ambiguous abnormal events are detected. Here we compare the Fusion and Parallel versions of DEDDnet [17], the Parallel version is similar to the proposed BiSP, i.e., it does not fuse the bidirectional prediction information and implements two independent frame prediction methods. Since the detection results at Sh-Tech are not provided for the parallel version, this paper compared to the fusion version, the BiSP increases by 1.9%. Although these two results are approximate, the ability of our method to learn features independently of forward and backward predictions indicates that our proposed method is more robust.

We further compare our method with state-of-the-art methods that use object-level images as inputs, as shown in Table 4. (1) These methods demonstrate strong performance, particularly those incorporating pose data [37–39], which perform better on the Sh-Tech dataset. This is because skeletal representations effectively eliminate redundant background information, allowing the model to focus on human-related anomaly behaviors while mitigating the risk of overfitting. (2) Without using object-level images as input, our method still achieves competitive performance on the ped2 dataset, attributed to the nature of ped2 as a single grayscale scene, which inherently minimizes background interference. (3) These methods [15,37,40,41] employ multi-task learning, combining tasks such as reconstruction, prediction, classification and segmentation, etc., compared to our use of only the prediction task. (4) Most importantly, the FPS of these methods that require preprocessing steps (object detection, optical flow) is much lower than that of VAD methods that rely solely on source video frames as input.

4.3. Model complexity and running time

Our model is implemented in PyTorch, and all tests are conducted on a single RTX 4090 GPU. The proposed BiSP has approximately 8.45M parameters and achieves a processing speed of 45 FPS, which includes bidirectional video frame generation and anomaly prediction. We also compare our model with other representative methods, including reconstruction-only (MemAE [6]) and prediction-only (MNAD [21]) methods. The summary of these results is presented in Table 5, our model's parameter count is second only to MemAE [6]. Despite using a dual-stream network for bidirectional prediction, we effectively control the number of channels in the network to achieve higher performance with fewer computational resources. In the inference phase, we use a multi-scale evaluation mode for anomaly prediction, achieving a

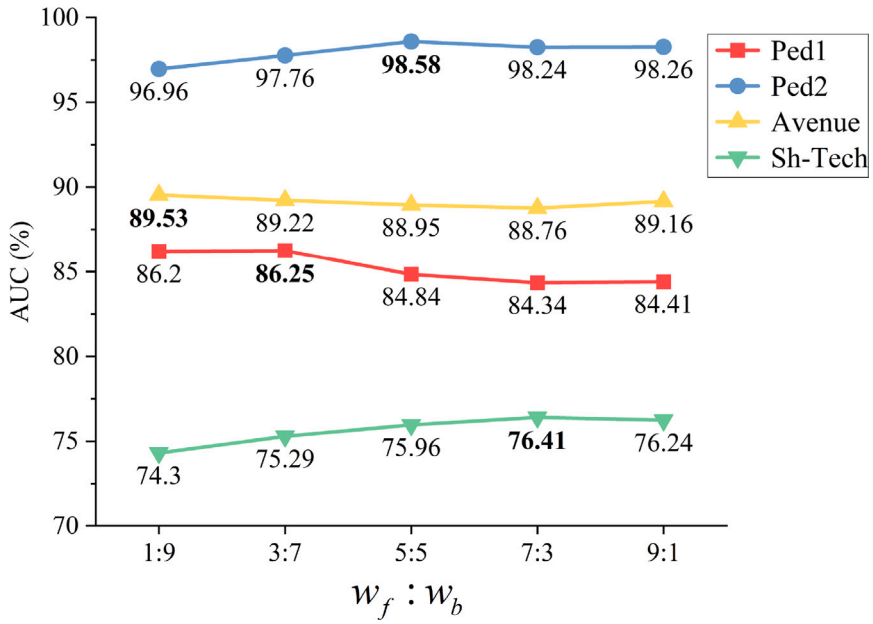

Fig. 4. Comparison of AUC for different error weight hyperparameters, bolded for optimal performance.

Table 5
The Parameters and FPS of our method and other VAD method. + indicates the result of the reimplementation of the method.

Method	Device (GPU)	Parameters (M)	FPS
MemAE [6] +	RTX 4090	6.50	31
MNAD [21] +	RTX 4090	15.00	87
Frame-Pred [12]	GTX TITAN	349.00	25
MSN-net [35]	GTX TITAN XP	9.16	95
BiSP(Ours)	RTX 4090	8.45	45 (75)

Table 6
Ablation studies of the proposed BiSP on Ped2 and Avenue datasets.

Model	SkipF	VarCA	ConSA	Ped2	Avenue
1	✗	✗	✗	97.0	85.0
2	✓	✗	✗	97.5	85.8
3	✓	✓	✗	98.2	87.0
4	✓	✗	✓	98.0	87.1
5	✗	✓	✓	98.0	87.3
6	✓	✓	✓	98.6	89.0

processing rate of 45 FPS. This performance meets real-time inference requirements for most videos. Without this evaluation mode, our model can reach 75 FPS, whereas MemAE achieves only 31 FPS due to its use of 3D convolutions.

4.4. Ablation study

4.4.1. Error weights hyperparameter analysis

We analyze the impact of different error weights hyperparameters on the AUC performance through experiments on four datasets. As shown in Fig. 4, we use five sets of hyperparameters to evaluate the performance of different error weights, where the horizontal axis represents the relationship between w_f and w_b in terms of proportions, i.e., $3 : 7$ means that w_f is 0.3 and w_b is 0.7. It can be intuitively observed that the AUC values across different error weights do not fluctuate much, and in particular, there is only a 0.78% fluctuation in Avenue. Although such fluctuations are acceptable in other datasets, how to choose appropriate error weight ratios for different datasets to optimize the AUC, is also a problem that needs to be considered by all hybrid methods.

4.4.2. Component and training strategy analysis

We conduct ablation studies to measure the effectiveness of skip-frame (SkipF) and the performance gained by each attention. The ablation study utilizes the Ped2 and Avenue datasets as an example and verifies the effectiveness by combining the proposed attentions one by one, and lastly Model 1 and Model 5 were tested to specifically validate SkipF. The results of the ablation studies, as summarized in Table 6, show the contribution of each module. To ensure the consistency of the ablation studies, the error weight hyperparameters w_f and w_b in this subsection were both set to 0.5.

It can be seen from Table 6, that the variant Models 2, 3, and 4 reduce the detection accuracy. The attention mechanism has less effect on ped2 due to its clear object and large boundary disparity, meanwhile, skip frames enhance the motion features of the objects. In contrast, in the more complex Avenue dataset, the effect of attention is more significant. As Model 6 shows, when adding two attentions in Model 2, the AUC performance significantly improved by 1.1% and 3.2%, respectively, and combining the results of Models 3 and 4 revealed that employing only single attention may cause abnormal generalization and reduce performance. Compared to Models 2 and 6, Models 1 and 5 use consecutive frames as model inputs instead of SkipF, resulting in a slight decrease in performance. This is because skip frames enhance the extraction efficiency of motion features. Anomaly events in the existing dataset are heavily associated with fast motion, and thus the model trained using the skip frame works best. Meanwhile, as shown in Model 1, using bidirectional prediction without any components still achieves better performance. This is due to the ability of the forward and backward prediction frames to catch the anomaly error simultaneously, which maximizes the detection efficiency.

Due to the extensive application of attention mechanisms in VAD, we conducted additional ablation studies on mainstream attention mechanisms. Table 7 shows the results of replacing the proposed VarCA and ConSA with different attention mechanisms. In Models 1, 2 and 3, traditional CA, Efficient Channel Attention (ECA) [42], and Residual Channel Attention (RCA) [34] replace VarCA, respectively. The results show that without variance attention, the AUC detection performance decreases across all three datasets. However, the decrease is smaller for Ped1, likely because it was captured by a camera perpendicular

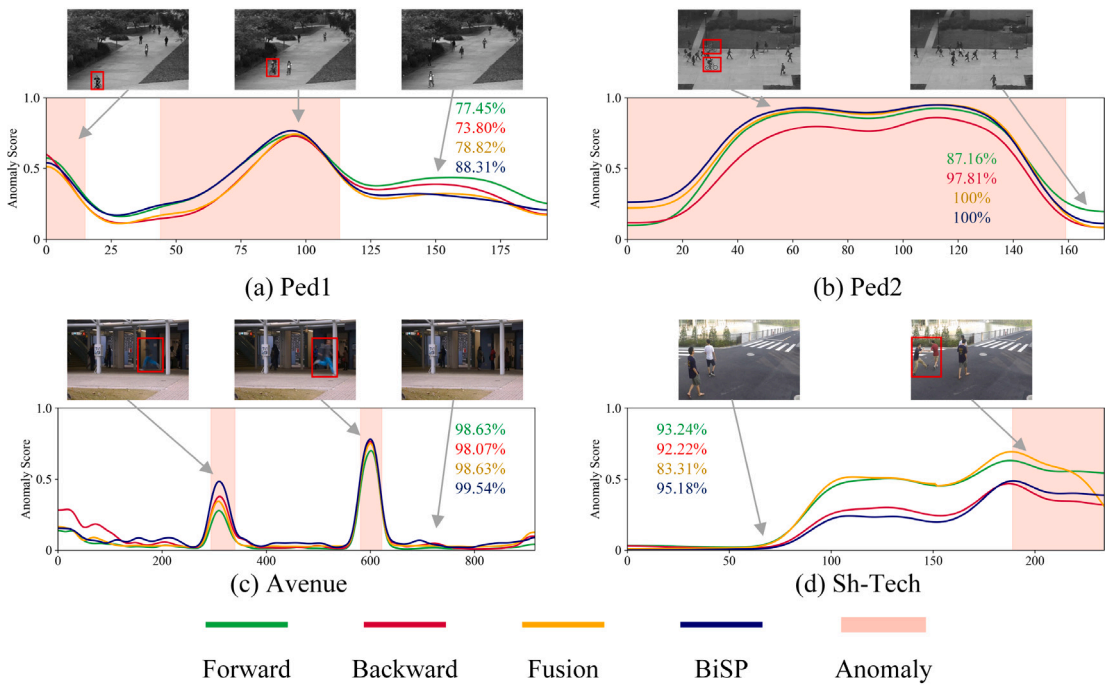

Fig. 5. Anomaly score curves and scores for several testing video clips of different methods on four benchmark datasets. The different color curves and numbers represent the score curves and scores of different models. The area marked by the red border contains abnormal events. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 7
Ablation studies of the attention mechanism on Ped1, Ped2, and Avenue datasets.

Model	Attention	Ped1	Ped2	Avenue
1	VarCA → CA	84.4	97.8	86.9
2	VarCA → ECA [42]	83.2	97.8	87.0
3	VarCA → RCA [34]	83.6	98.1	87.3
4	ConSA → SA	83.0	97.7	87.3
5	ConSA → FSA [11]	82.3	96.5	85.3
6	VarCA/ConSA → CBAM [28]	82.3	96.8	86.5

Table 8
Frame-level AUC comparison of anomaly scores under different feature learning methods.

	Ped1	Ped2	Avenue	SH-Tech
Forward	82.9	96.3	86.4	73.6
Backward	83.4	96.6	86.8	73.4
Fusion	83.9	98.0	88.5	74.6
BiSP	84.8	98.6	89.0	76.0

to the road, leading to less pronounced target movement. In contrast, the cameras for Ped2 and Avenue are positioned parallel to the road, capturing more pronounced motion. In Models 4 and 5, traditional SA and Fusion Spatial Attention (FSA) [11] replace ConSA. To adapt FSA, certain modules are modified. Specifically, skip connection and up-sampling feature extraction are fused and used as input features for FSA. Lacking contextual features, this variant model loses the ability to capture target features across different scales. Specifically, the FSA [11] fuses skip connection features early, leading to missing features in the subsequent decoding process. Model 6 uses CBAM [28], widely applied in computer vision, to replace the proposed VarCA and ConSA. Although CBAM is effective in various tasks, using multiple CBAMs in Model 6 hampers effective feature extraction.

Furthermore, to explore the effectiveness of the proposed method under different training strategies, this paper also designs three additional ablation models, i.e., "Forward", "Backward" and "Fusion". Table 8 presents the AUC performance of these training strategies across various datasets. "Forward" and "Backward" denote models that use of forward or backward skip frames, respectively, for training and testing only in the framework of single AE. "Fusion" builds on the BiSP framework by transforming the input features of the decoder. Instead of using only the separate output features from the forward or backward encoder, the "Fusion" combines the low-dimensional latent space features from dual-AE. Here, we use simple element-wise addition to represent the feature fusion process. For model evaluation and AUC calculation, the single AE model calculates the error solely from the corresponding forward or backward frames. In contrast, the dual-AE model predicts the same frames from both the forward and backward directions during testing, so the average of the two predicted frames is used to calculate the error.

From Table 8 shows that the AUC of the "Forward" and "Backward" variant models for a single AE is lower than that of the dual-AE model, suggesting that the bidirectional training strategy outperforms the unidirectional. Moreover, the AUC of the "Fusion" model is lower than the proposed method, contradicting the conclusion drawn in DEDDnet [17]. This is because DEDDnet uses an identical video frame for bidirectional training, while the prediction frame does not overlap with the training video frame. In contrast, the bidirectional training video frames of BiSP do not overlap and the forward training video frames contain the prediction frame of the backward AE, as does the backward training. This cross-fusion disrupts the bidirectional feature extraction process, resulting in a minor performance drop.

To specifically analyze the discrimination ability of the different models for anomaly and normal, Fig. 5 shows its anomaly curves and scores in the testing video clips. Anomaly scores are obtained based on different thresholds, which depend on the false and true positive rates. We categorized each testing video frame by normalized anomaly scores and thresholds, i.e., a score close to 0 indicates normal, and close to 1 indicates an anomaly. The anomaly in the testing video frames in Fig. 5 is more obvious, so the curves trend similarly for the different methods, but there is still a slight difference. For instance, the dual-AE model in Fig. 5(a) has lower scores in the normal interval than the single AE model, while performing higher in the anomaly interval, which further

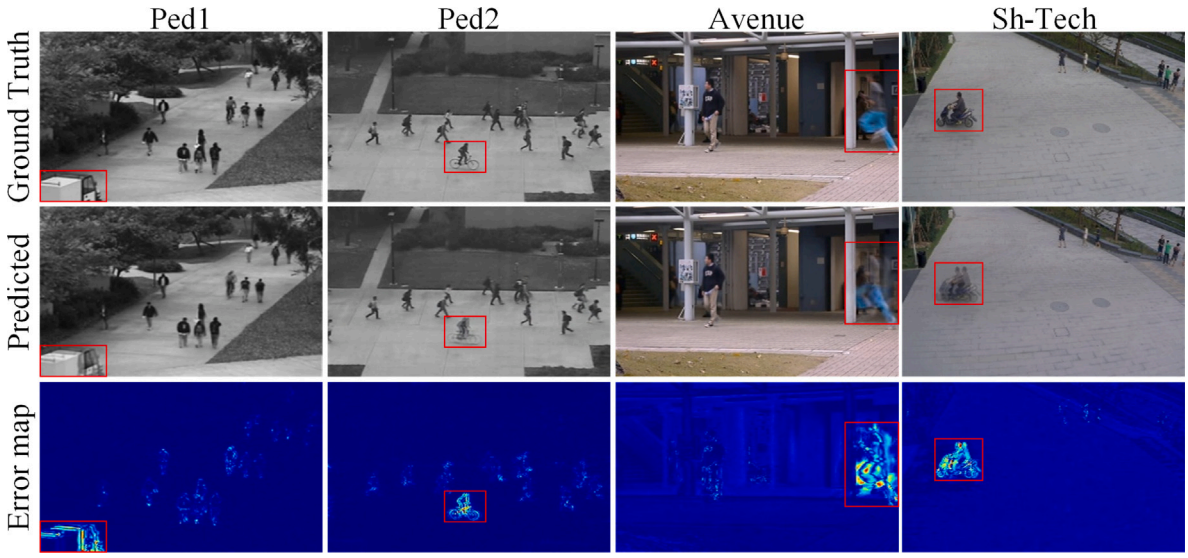

Fig. 6. Visualization results for four datasets: Ground Truth (top), Predicted frames (middle), and Error maps (bottom), where brighter colors represent larger errors. The area marked by the red border contains abnormal events. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

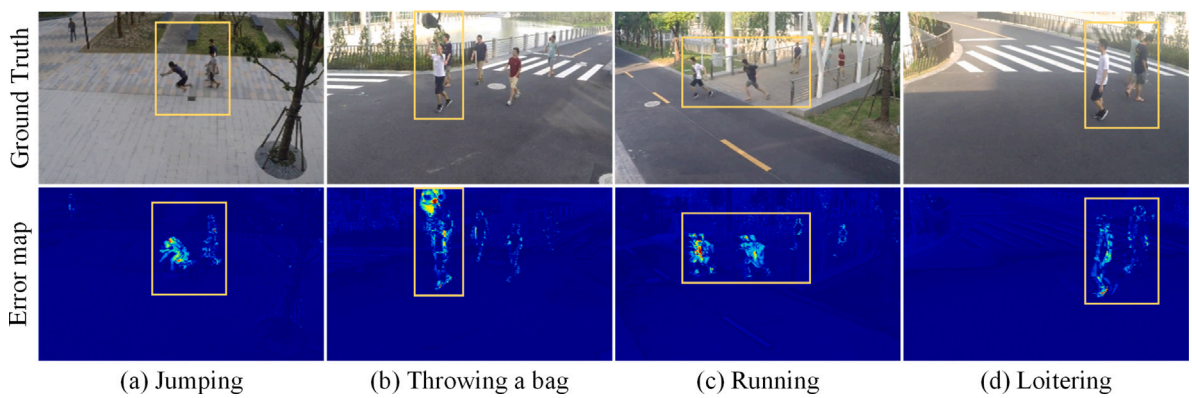

Fig. 7. Examples of ambiguous boundary in the Sh-Tech dataset: Ground Truth (top) and Error maps (bottom). The area marked by the yellow border contains normal and abnormal events. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

demonstrates the advantage of bidirectional prediction. In Fig. 5(d), the plunge of the fusion model's score in the anomaly interval is caused by the cross information interfering with the prediction model, which results in a much lower performance than the proposed method.

4.5. Visualization results

Fig. 6 shows the visualization results of the proposed BiSP in different datasets and marks the exact location of the anomalies. It is evident that, due to the bidirectional prediction, anomalous events exhibit overlapping blurred states. This effect is particularly pronounced for anomalies with significant boundaries, where the brightness of the prediction error maps is markedly higher. In contrast, the errors for normal events are negligible. In the Avenue dataset, in addition to the obvious predictions that can be observed, some of the backgrounds are found to have slight errors compared to the surrounding information. Although Avenue belongs to a single scene, it can find the phenomenon of background bias by observing the training set, which shows different degrees of light intensities in various video clips.

In contrast to most examples of the clear boundaries shown in Figs. 6, 7 illustrates the ambiguous boundary in the Sh-Tech dataset, including four anomaly events, i.e., jumping, throwing a bag, running, and loitering. It can be seen that ambiguous abnormal events are distinguished only by human behavior. In the error maps, the proposed BiSP achieves efficient discrimination of the ambiguous anomalies in Fig. 7(a)–(c), with a significant gap between normal and anomaly. For the loitering behavior in Fig. 7(d), the white-dressed human is much less morphological differences from the others, such anomalies in this behavior are more difficult to detect. Still, BiSP achieves the desired discriminative ability by widening the intra-domain disparity between different events.

5. Discussion

In this section, we discuss the limitations of our method from the perspective of unsupervised VAD methods, datasets, and tasks. Despite achieving significant results by our proposed BiSP in existing unsupervised datasets, our method processes the entire image frame as input, which may pose the potential risk of being difficult to capture for tiny anomalous targets as compared to object-level image. Moreover, most existing VAD datasets are single-modal, and the detection results often lack explainability. This leads current VAD tasks to be biased towards video background reconstruction rather than video content understanding. Additionally, how to consider scene-dependent anomalies instead of limiting to stereotypical anomaly attributes is a major challenge for future VAD. For instance, future methods should have different

definitions and explainability for the same behavior in different scenes. Therefore, future work will explore Large Vision-Language Models (LVLMs) to focus on both visual and textual information across diverse scenes, thus enhancing anomaly discrimination and hopefully providing explainability.

6. Conclusion

In this paper, we propose a bidirectional skip-frame prediction method for unsupervised VAD, called BiSP, which enhances detection performance through the fusion error of forward and backward prediction. The proposed BiSP leverages different input strategies and two attention mechanisms to widen the discrimination boundary between normal and abnormal events. Based on a dual-stream autoencoder, we construct the variance channel attention with the parallel structure and the context spatial attention with the serial structure. These mechanisms improve the feature representation of normal events by increasing the discrimination boundary of different events from motion patterns and target sizes. The VarCA mechanism can record the motion patterns of normal events, thus enabling a more sensitive capture of abnormal motion. The ConSA mechanism focuses on the different object scales, allowing the model to comprehensively detect various anomalies. Experimental results on benchmark datasets and the ablation study validate the performance of our method as superior or equivalent to state-of-the-art methods. In future work, we aim to focus on semantic-level or object-level information for different scenes through LVLM, which guarantees, at least in principle, that the diverse anomalies are better defined.

CRediT authorship contribution statement

Jiahao Lyu: Writing – original draft, Visualization, Methodology, Formal analysis, Conceptualization. **Minghua Zhao:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition. **Jing Hu:** Writing – review & editing, Supervision, Project administration, Funding acquisition. **Runtao Xi:** Writing – review & editing, Validation, Investigation, Data curation. **Xuewen Huang:** Writing – review & editing, Visualization, Validation, Resources, Investigation. **Shuangli Du:** Writing – review & editing, Supervision, Project administration, Funding acquisition. **Cheng Shi:** Writing – review & editing, Visualization, Project administration. **Tian Ma:** Writing – review & editing, Visualization, Supervision, Resources.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This research was funded by Natural Science Foundation of Shaanxi Province, China (2024JC-ZDXM-35, 2024JC-YBMS-458, 2024JC-YBMS-573), National Natural Science Foundation of China (No. 52275511) and Young Talent Fund of Association for Science and Technology in Shaanxi, China (20240146).

Data availability

Data will be made available on request.

References

[1] Y. Gu, Y. Lin, K.-T. Cheng, H. Chen, Revisiting deep ensemble uncertainty for enhanced medical anomaly detection, in: International Conference on Medical Image Computing and Computer-Assisted Intervention, MICCAI, Springer, 2024, pp. 520–530.
[2] Z. Fang, X. Wang, H. Li, J. Liu, Q. Hu, J. Xiao, Fastrecon: Few-shot industrial anomaly detection via fast feature reconstruction, in: Proceedings of the IEEE/CVF International Conference on Computer Vision, 2023, pp. 17481–17490.
[3] D. Li, X. Nie, R. Gong, X. Lin, H. Yu, Multi-branch GAN-based abnormal events detection via context learning in surveillance videos, IEEE Trans. Circuits Syst. Video Technol. 34 (5) (2023) 3439–3450.
[4] R.T. Ionescu, F.S. Khan, M.-I. Georgescu, L. Shao, Object-centric auto-encoders and dummy anomalies for abnormal event detection in video, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2019, pp. 7842–7851.
[5] J. Chen, L. Li, L. Su, Z.-j. Zha, Q. Huang, Prompt-enhanced multiple instance learning for weakly supervised video anomaly detection, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2024, pp. 18319–18329.
[6] D. Gong, L. Liu, V. Le, B. Saha, M.R. Mansour, S. Venkatesh, A.v.d. Hengel, Memorizing normality to detect anomaly: Memory-augmented deep autoencoder for unsupervised anomaly detection, in: Proceedings of the IEEE/CVF International Conference on Computer Vision, 2019, pp. 1705–1714.
[7] J.-C. Feng, F.-T. Hong, W.-S. Zheng, Mist: Multiple instance self-training framework for video anomaly detection, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2021, pp. 14009–14018.
[8] S. Majhi, R. Dai, Q. Kong, L. Garattoni, G. Francesca, F. Brémond, OE-CTST: Outlier-embedded cross temporal scale transformer for weakly-supervised video anomaly detection, in: Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision, 2024, pp. 8574–8583.
[9] G. Pang, C. Shen, L. Cao, A.V.D. Hengel, Deep learning for anomaly detection: A review, ACM Comput. Surv. 54 (2) (2021) 1–38.
[10] C. Park, M. Cho, M. Lee, S. Lee, FastAno: Fast anomaly detection via spatio-temporal patch transformation, in: Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision, 2022, pp. 2249–2259.
[11] X. Zhang, J. Fang, B. Yang, S. Chen, B. Li, Hybrid attention and motion constraint for anomaly detection in crowded scenes, IEEE Trans. Circuits Syst. Video Technol. 33 (5) (2022) 2259–2274.
[12] W. Liu, W. Luo, D. Lian, S. Gao, Future frame prediction for anomaly detection–a new baseline, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2018, pp. 6536–6545.
[13] J. Liang, Y. Xiao, J.T. Zhou, F. Yang, T. Li, Z. Fang, C 2 Net: content-dependent and-independent cross-attention network for anomaly detection in videos, Appl. Intell. 54 (2) (2024) 1980–1996.
[14] Z. Fang, J. Liang, J.T. Zhou, Y. Xiao, F. Yang, Anomaly detection with bidirectional consistency in videos, IEEE Trans. Neural Netw. Learn. Syst. 33 (3) (2020) 1079–1092.
[15] Z. Liu, Y. Nie, C. Long, Q. Zhang, G. Li, A hybrid video anomaly detection framework via memory-augmented flow reconstruction and flow-guided frame prediction, in: Proceedings of the IEEE/CVF International Conference on Computer Vision, 2021, pp. 13588–13597.
[16] W. Luo, W. Liu, D. Lian, S. Gao, Future frame prediction network for video anomaly detection, IEEE Trans. Pattern Anal. Mach. Intell. 44 (11) (2021) 7505–7520.
[17] Y. Zhong, X. Chen, Y. Hu, P. Tang, F. Ren, Bidirectional spatio-temporal feature learning with multiscale evaluation for video anomaly detection, IEEE Trans. Circuits Syst. Video Technol. 32 (12) (2022) 8285–8296.
[18] Y. Zhang, X. Fang, F. Li, L. Yu, Mutual learning inspired prediction network for video anomaly detection, in: Chinese Conference on Pattern Recognition and Computer Vision, PRCV, Springer, 2022, pp. 578–593.
[19] D. Lappas, V. Argyriou, D. Makris, Dynamic distinction learning: adaptive pseudo anomalies for video anomaly detection, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2024, pp. 3961–3970.
[20] R. Kommanduri, M. Ghorai, DAST-Net: Dense visual attention augmented spatio-temporal network for unsupervised video anomaly detection, Neurocomputing (2024) 127444.
[21] H. Park, J. Noh, B. Ham, Learning memory-guided normality for anomaly detection, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2020, pp. 14372–14381.
[22] S. Qiu, J. Ye, J. Zhao, L. He, L. Liu, E. Bicong, X. Huang, Video anomaly detection guided by clustering learning, Pattern Recognit. 153 (2024) 110550.
[23] C. Huang, J. Wen, Y. Xu, Q. Jiang, J. Yang, Y. Wang, D. Zhang, Self-supervised attentive generative adversarial networks for video anomaly detection, IEEE Trans. Neural Netw. Learn. Syst. 34 (11) (2022) 9389–9403.
[24] Z. Ning, Z. Wang, Y. Liu, J. Liu, L. Song, Memory-enhanced appearance-motion consistency framework for video anomaly detection, Comput. Commun. 216 (2024) 159–167.

[25] X. Huang, Y. Hu, X. Luo, J. Han, B. Zhang, X. Cao, Boosting variational inference with margin learning for few-shot scene-adaptive anomaly detection, IEEE Trans. Circuits Syst. Video Technol. 33 (6) (2022) 2813–2825.
[26] Y. Hao, J. Li, N. Wang, X. Wang, X. Gao, Spatiotemporal consistency-enhanced network for video anomaly detection, Pattern Recognit. 121 (2022) 108232.
[27] L. Wang, J. Tian, S. Zhou, H. Shi, G. Hua, Memory-augmented appearance-motion network for video anomaly detection, Pattern Recognit. 138 (2023) 109335.
[28] S. Woo, J. Park, J.-Y. Lee, I.S. Kweon, Cbam: Convolutional block attention module, in: Proceedings of the European Conference on Computer Vision, ECCV, 2018, pp. 3–19.
[29] S. Wang, Z. Miao, Anomaly detection in crowd scene, in: IEEE 10th International Conference on Signal Processing Proceedings, IEEE, 2010, pp. 1220–1223.
[30] C. Lu, J. Shi, J. Jia, Abnormal event detection at 150 fps in matlab, in: Proceedings of the IEEE International Vonference on Vomputer Vision, 2013, pp. 2720–2727.
[31] W. Luo, W. Liu, S. Gao, A revisit of sparse coding based anomaly detection in stacked rnn framework, in: Proceedings of the IEEE International Conference on Computer Vision, 2017, pp. 341–349.
[32] Y. Chang, Z. Tu, W. Xie, J. Yuan, Clustering driven deep autoencoder for video anomaly detection, in: Proceedings of the European Conference on Computer Vision, ECCV, Springer, 2020, pp. 329–345.
[33] Z. Yang, P. Wu, J. Liu, X. Liu, Dynamic local aggregation network with adaptive clusterer for anomaly detection, in: Proceedings of the European Conference on Computer Vision, ECCV, Springer, 2022, pp. 404–421.
[34] V.-T. Le, Y.-G. Kim, Attention-based residual autoencoder for video anomaly detection, Appl. Intell. 53 (3) (2023) 3240–3254.
[35] Y. Liu, D. Li, W. Zhu, D. Yang, J. Liu, L. Song, MSN-net: Multi-scale normality network for video anomaly detection, in: ICASSP 2023-2023 IEEE International Conference on Acoustics, Speech and Signal Processing, ICASSP, IEEE, 2023, pp. 1–5.
[36] C. Park, D. Kim, M. Cho, M. Kim, M. Lee, S. Park, S. Lee, Fast video anomaly detection via context-aware shortcut exploration and abnormal feature distance learning, Pattern Recognit. (ISSN: 0031-3203) (2024) 110877.
[37] A. Barbalau, R.T. Ionescu, M.-I. Georgescu, J. Dueholm, B. Ramachandra, K. Nasrollahi, F.S. Khan, T.B. Moeslund, M. Shah, SSMTL++: Revisiting self-supervised multi-task learning for video anomaly detection, Comput. Vis. Image Underst. 229 (2023) 103656.
[38] O. Hirschorn, S. Avidan, Normalizing flows for human pose anomaly detection, in: Proceedings of the IEEE/CVF International Conference on Computer Vision, 2023, pp. 13545–13554.
[39] X. Zeng, Y. Jiang, W. Ding, H. Li, Y. Hao, Z. Qiu, A hierarchical spatio-temporal graph convolutional neural network for anomaly detection in videos, IEEE Trans. Circuits Syst. Video Technol. 33 (1) (2021) 200–212.
[40] M.-I. Georgescu, A. Barbalau, R.T. Ionescu, F.S. Khan, M. Popescu, M. Shah, Anomaly detection in video via self-supervised and multi-task learning, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2021, pp. 12742–12752.
[41] M.I. Georgescu, R.T. Ionescu, F.S. Khan, M. Popescu, M. Shah, A background-agnostic framework with adversarial training for abnormal event detection in video, IEEE Trans. Pattern Anal. Mach. Intell. 44 (9) (2021) 4505–4523.
[42] Q. Wang, B. Wu, P. Zhu, P. Li, W. Zuo, Q. Hu, ECA-Net: Efficient channel attention for deep convolutional neural networks, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2020, pp. 11534–11542.

Jiahao Lyu is a Ph.D. student at the School of Computer Science and Engineering, Xi'an University of Technology. He received his M.S. degree from the School of Computer Science and Technology, Xi'an University of Science and Technology in 2023. His research interests include computer vision, video anomaly detection, and reinforcement learning.

Minghua Zhao received the Ph.D. degree from Sichuan University, Chengdu, China, in 2006. She is currently with the School of Computer Science and Engineering, Xian University of Technology, Xian, China. Her research interests include digital image processing, pattern recognition, computer graphics, and computer vision.

Jing Hu received her Ph.D. degree in Communication and Information Systems from Xidian University, Xi'an, China, in 2018. She is currently an Associate Professor at the School of Computer Science and Engineering, Xi'an University of Technology, Xi'an, China. Her research interests include hyperspectral image super-resolution and anomaly detection.

Runtao Xi obtained his Bachelors Degree in Software Engineering from LangFang Normal University in 2019 and his Masters Degree in Software Engineering from Xian University of Science and Technology in 2022. Currently, he is working at Changzhou Research Institute on computer vision engineering research and software engineering. His research interests include computer vision and reinforcement learning.

Xuewen Huang received the B.S. degree in Software Engineering from Inner MongoliaAgricultural University in 2021 and is currently pursuing a M.S. degree in Computer Science and Technology at the School of Computer Science and Engineering, Xi'an University of Technology. His research interests include computer vision, video anomaly detection.

Shuangli Du received the Ph.D. degrees in computer science from Sichuan University in 2017. In 2014, she was a Visiting Student with the Agency for Science, Technology and Research (A*STAR), Singapore. Her research interests include low-rank decomposition, computer vision, and pattern recognition.

Cheng Shi received the B.S. degree from Xian University of Architecture and Technology, Shanxi, China, in 2009, the M.S. and ph.D. degrees from Xidian University, Shanxi, China, in 2012 and 2016, respectively, and worked as the postdoctoral research fellow with University of Macau, China, from 2016 to 2018.

She is currently working as an Associate Professor with the School of Computer Science and Engineering, Xian University of Technology, China. She is the author or coauthor of more than 20 publications on image processing, including international journal papers, conference papers, and book chapters. She has also been a reviewer of several journals, such as IEEE TGRS, IEEE JSTARS, IEEE TNNLS, and MTAPs.

Her research interests include deep learning, image processing, and pattern recognition.

Tian Ma was born in Henan, China, in 1982. He received the B.S. degree in measurement and control technology and instrument, the M.S. degree in software engineering, and the Ph.D. degree in information and communication engineering from Northwestern Polytechnical University, Xian, China, in 2003, 2006, and 2011, respectively. Since 2014, he has been an Professor with the College of Computer Science and Technology, Xian University of Scienceand Technology, China. He is the author of more than 20 papers, and more than ten software copyright and invention patent. His research interests include image processing, computer graphics, and 3D simulation and visualization. He has served as a Conference Co-Chair for Seventh International Conference on Image Vision and Computing (ICIVC 2022), an Organizing Committee Co-Chair for 14th International Conference on Verification and Evaluation of Computer and Communication Systems (VECoS 2020), and a Local Arrangement Chair for IEEE International Conference on Signal Processing, Communication and Computing (ICSPCC 2022).